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Editor-in-Chief
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Dear Editor,

We would like to submit our manuscript entitled “*Adaptive Multi-Representation Fusion via Dual-Channel PCNN with Multi-Scale Convolution for Vibration Signal Fault Diagnosis*” for consideration for publication in *IEEE Transactions on Instrumentation and Measurement*.

Vibration signal analysis plays a critical role in the fault diagnosis of rotating machinery. However, in practical industrial environments, vibration signals exhibit pronounced non-stationarity and complex time-varying characteristics, which significantly limit the effectiveness of conventional single-representation diagnostic methods. Existing approaches based on either time–frequency representations or temporal encodings alone capture only partial aspects of the rich information embedded in vibration signals, and the complementary nature of heterogeneous representations remains largely underexploited. Moreover, current fusion strategies typically rely on fixed weighting or naive concatenation, lacking the adaptivity to dynamically regulate the relative contributions of different representations on a per-sample or per-region basis.

To address these limitations, this work proposes a two-stage framework: an adaptive weighted dual-channel pulse-coupled neural network (AW-DPCNN) for multi-representation fusion, and a multi-scale convolution and channel attention enhanced VGG16 (MSCA-VGG16) classifier for robust fault identification. The AW-DPCNN adaptively integrates short-time Fourier transform (STFT) spectrograms and Gramian angular difference field (GADF) images through contrast-guided dynamic weighting. The MSCA-VGG16 further exploits the fused representations via multi-scale convolution. The main contributions of this work include:

1. An AW-DPCNN fusion mechanism that adaptively integrates heterogeneous time–frequency and temporal-encoded representations, introducing contrast-guided adaptive weighting to dynamically regulate per-channel contributions and dual-channel external stimuli with representation-specific convolutional kernels for feature reinforcement.

2. An MSCA-VGG16 classification network that incorporates multi-scale convolution for capturing fault patterns at multiple temporal and spectral granularities, squeeze-and-excitation channel attention for dynamic frequency-band recalibration, and a compact embedding head for enhanced discriminability.
3. Extensive experiments on the CWRU bearing dataset, including backbone comparisons, comprehensive ablation studies, representation comparisons across twelve time-frequency and temporal encoding combinations, and noise robustness evaluations, systematically validate each component of the proposed framework. Cross-sensor generalization to the fan-end position (97.45%) and cross-sampling-rate robustness at 48 kHz (94.81%) are further demonstrated.

Experimental results on the CWRU bearing dataset demonstrate that the proposed framework achieves 99.85% accuracy on the 12 kHz drive-end benchmark, outperforming seven representative backbone architectures. Ablation studies confirm that the adaptive weighting mechanism contributes 2.93% improvement over single-representation baselines, while channel attention alone adds 2.67%. Noise robustness evaluations validate stable performance under degraded signal conditions, and generalization experiments confirm the robustness of the learned representations across sensor positions and sampling rates without dataset-specific tuning.

We believe that this work fits well within the scope of *IEEE Transactions on Instrumentation and Measurement*, particularly in the areas of intelligent signal processing, sensor-based fault diagnosis, and deep learning for instrumentation and measurement applications. This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the submission and agree with its contents.

Thank you for your time and consideration.

Sincerely,

Zonglong Bai

On behalf of all co-authors

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